Tide retrospective: one temperature determines the quartic coefficient

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The recovery thread so far identifies the leading data of a pure even-monomial potential from partition-function asymptotics. The germbij note's full Section 7.4 recovers subleading coefficients too, by an inductive expansion pairing; formalising that induction is a multi-tide programme. Deliberation identified a minimal step that is not a fragment of the induction but a complementary mechanism: for a *nonnegative* perturbation, the partition function is strictly monotone in the coefficient, so the subleading coefficient is already determined by Z at a *single* temperature, no asymptotics needed.

2 The thing formalised

Four declarations in `Laplace/OneD/RecoveryMonotone.lean`, sorry-free with zero warnings, organized as the consult's ladder. The measure-theoretic core[1]: if $L_1 \le L_2$ pointwise, e^{-tL_1} is integrable, and $\{L_1 < L_2\}$ has positive measure, then $Z_{L_2}(t) < Z_{L_1}(t)$ for $t > 0$. The topological corollary[2]: continuity plus a single point with $L_1(x_0) < L_2(x_0)$ suffices. Strict antitonicity[3]: for continuous V and continuous $g \ge 0$ positive somewhere, $b \mapsto Z_{V+bg}(t)$ is strictly decreasing on $b \ge 0$. And the recovery statement[4]: for $x^2/2 + b\,x^4$ with $b \ge 0$, equality of partition functions at one temperature forces equality of the coefficients.

3 Proof strategy

The core is three Mathlib moves: domination transfers integrability from e^{-tL_1} to e^{-tL_2}; the difference of exponentials is nonnegative with support containing $\{L_1 < L_2\}$, so the support-positivity lemma[5] converts positive measure of the strict locus into strict positivity of the integral; and `integral_sub` turns that into the strict inequality of partition functions. The corollary produces the positive-measure locus from `isOpen_lt` and `IsOpen.measure_pos`; antitonicity is the corollary applied to $V + b_1 g \le V + b_2 g$; recovery is `StrictAntiOn.injOn` against the seabed's monomial integrability.

[1] `partitionFunction_lt_of_le_of_measure_lt_pos`.
[2] `partitionFunction_lt_of_le_of_lt`.
[3] `partitionFunction_perturb_strictAntiOn`.
[4] `quartic_coefficient_recovery`.
[5] `integral_pos_iff_support_of_nonneg`.

4 Roadblocks and resolutions

Three build iterations. An `AEStronglyMeasurable` witness built by composing continuity lemmas elaborates as `rexp` composed with a lambda, which the rewrite inside `Integrable.mono'` cannot match against the beta-reduced integrand; stating the witness as an explicitly-typed `have` discharged by `fun_prop` fixes it (the same anchor discipline as the previous tide's constant-limit witness). And `nlinarith` needed its product hints ($0 \le t\, b_1\, g(x)$ and kin) spelled out at three sites.

5 Process note

A first draft of the tide log's numerical check quoted plausible invented values before the script had been run; the actual run gave different numbers. The corrected log records both. This is the third occurrence of this failure mode in the arc's history and the reason the check is required to be executed, not narrated.

6 What was Mathlib, what was new

Mathlib: everything in the core (domination, support positivity, integral subtraction, open-set positivity). Seabed: monomial integrability. New: the observation that order-theoretic recovery complements the note's expansion pairing, giving a stronger conclusion (one temperature) under a stronger hypothesis (signed perturbation).

7 Follow-ups

- The genuine expansion-based induction (recovering a_j of a general analytic potential from correction orders) remains the long-term recovery item; its first Lean step needs an expansion of Z for a perturbed monomial, which is anharmonic-programme machinery rather than a small tide.
- Tag the note's Section 7.4 with the recovery theorem once merged, with a remark that the Lean statement uses the monotonicity mechanism.